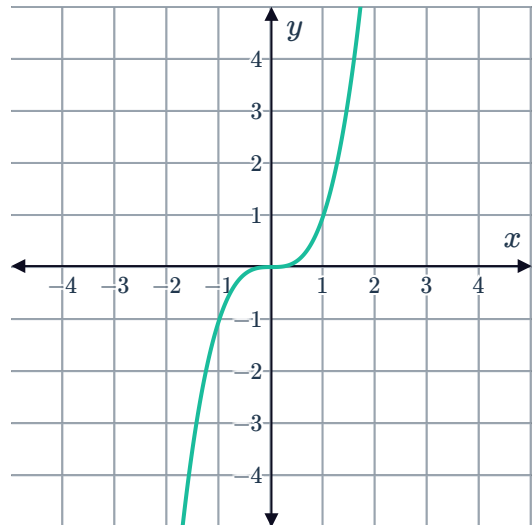


The shape of a cubic



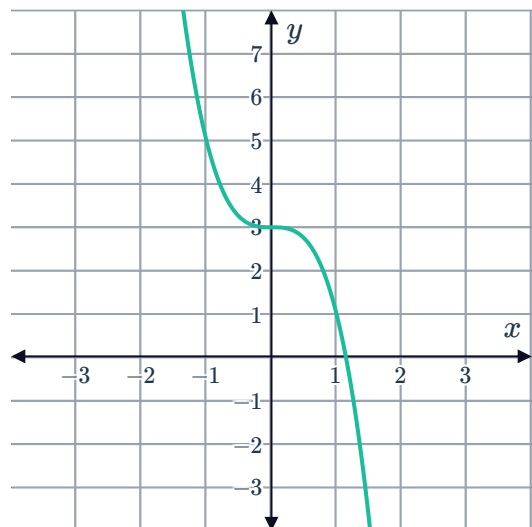
1 Consider the graph of $y = x^3$.

- a As x becomes larger in the positive direction (ie x approaches infinity), what happens to the corresponding y -values?
- b As x becomes larger in the negative direction (ie x approaches negative infinity), what happens to the corresponding y -values?



2 Consider the given graph of a cubic function.

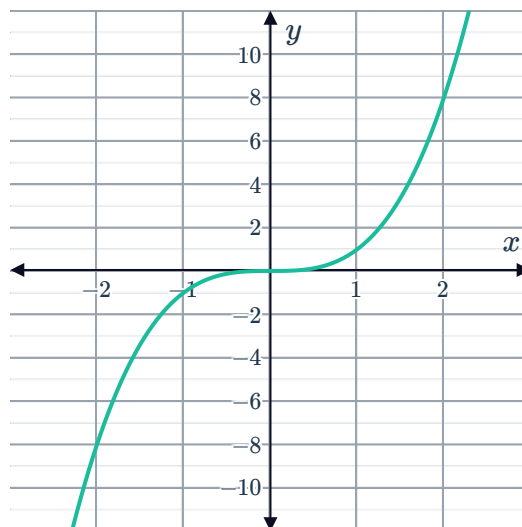
- a Determine whether the cubic is positive or negative.
- b State the coordinates of the y -intercept.
- c State the equation of the function.



3 Consider the cubic function $y = \frac{1}{2}x^3 + x$.

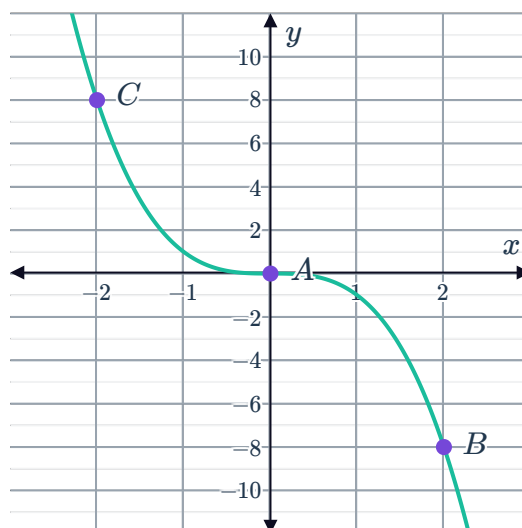
- a Determine whether the cubic is positive or negative.
- b Sketch the graph of $y = \frac{1}{2}x^3 + x$.
- c State the coordinates of the x -intercept.

- 4 Consider the graph of the function $y = x^3$.
Determine the point where the curve changes from being concave down to being concave up.
This is called the point of inflection.



- 5 Consider the graph of the function $y = -x^3$.
Out of the points A , B and C :

- a At which point is the curve concave up?
- b Which is the point of inflection?
- c At which point is the curve concave down?

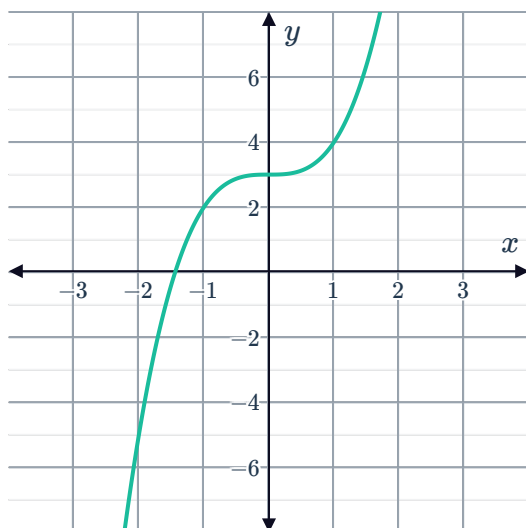


6 For each of the following functions:

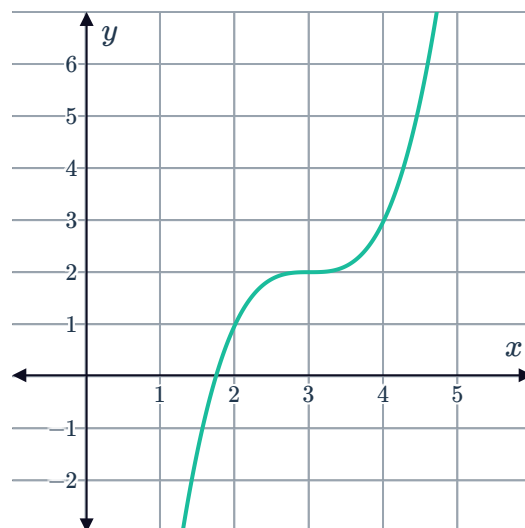


- i For what values of x is the cubic concave up?
- ii For what values of x is the cubic concave down?
- iii State the coordinates of the point of inflection.

a



b

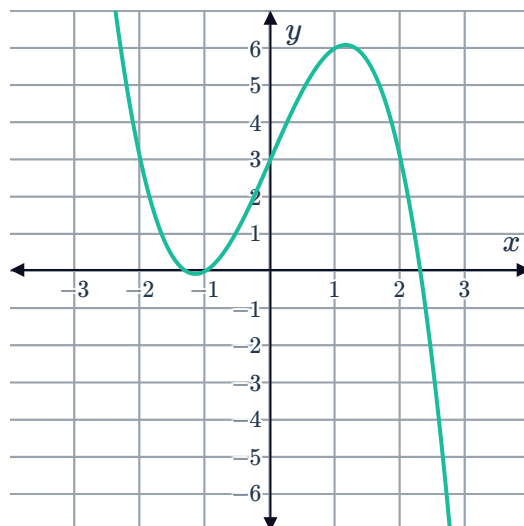


7 A cubic function is defined as $y = \frac{1}{2}x^3 + 4$.

- a Find the x -intercept of the function.
- b Find the y -intercept of the function.

8 Consider the graph of the function:

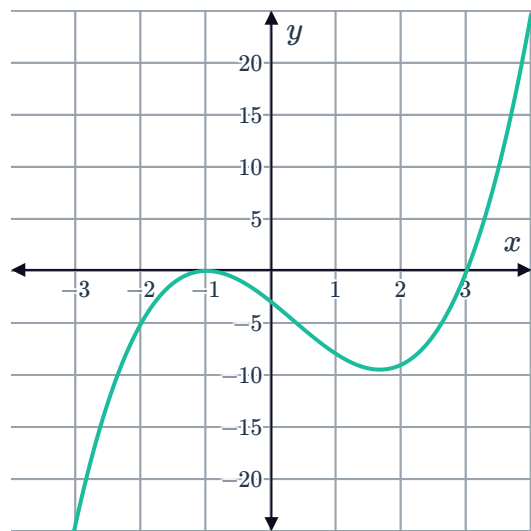
- a The equation of the function can be written as $y = ax^3 + bx^2 + cx + d$. Is the value of a positive or negative?
- b State the coordinates of the y -intercept.
- c For which values of x is the graph concave up?
- d For which values of x is the graph concave down?
- e State the coordinates of the point of inflection.



- 9 Consider the graph of the cubic function shown.



For what values of x is $y \geq 0$?



Transformations

- 10 Determine whether the following statements are true of the graphs of $y = \frac{1}{2}x^3$ and $y = x^3$.
- a One is a reflection of the other about the y -axis.
 - b y increases more rapidly on $y = \frac{1}{2}x^3$ than on $y = x^3$.
 - c $y = \frac{1}{2}x^3$ is a horizontal shift of $y = x^3$.
 - d y increases more slowly on $y = \frac{1}{2}x^3$ than on $y = x^3$.
- 11 The graph of $y = x^3$ has a point of inflection at $(0, 0)$. By considering the transformations that have taken place, find the point of inflection of each cubic curve below:
- a $y = \frac{2}{3}x^3$
 - b $y = x^3 + 3$
 - c $y = -x^3 + 4$

- 12 For each cubic function below:

- i Complete a table of values of the form:
- ii Sketch the graph.

x	-2	-1	0	1	2
y					

- a $y = x^3$
- b $y = x^3 - 2$
- c $y = -x^3 + 5$

13 Consider the curve $y = x^3 - 8$.



- a Find the x -intercept.
- b Find the y -intercept.
- c Find the horizontal point of inflection.
- d Sketch the graph of the curve.

14 Consider the curve $y = 3x^3 + 3$.

- a Find the x -intercept.
- b Find the y -intercept.
- c State the coordinates of the point of inflection.
- d Sketch the graph of the curve.

15 Consider the equation $y = x^3 - 3$.

- a Complete the following set of points for the given equation.
 $A(-3, \quad), B(-2, \quad), C(-1, \quad), D(0, \quad), E(1, \quad), F(2, \quad), G(3, \quad)$
- b Sketch the curve that results from the entire set of solutions for the equation being graphed.

16 Consider the cubic function $y = x^3 - 4$.

- a State the y -intercept of the function.
- b Complete the following table of values.

x	-1	1	2
y			

- c Find the domain of the function in interval notation.
- d Find the range of the function in interval notation.
- e Sketch the graph.

17 Bianca completed the table of values for the equation $y = 20 - x^3$.

x	-6	-4	-2	0	2	4
y	236	84	28	20	-4	-44

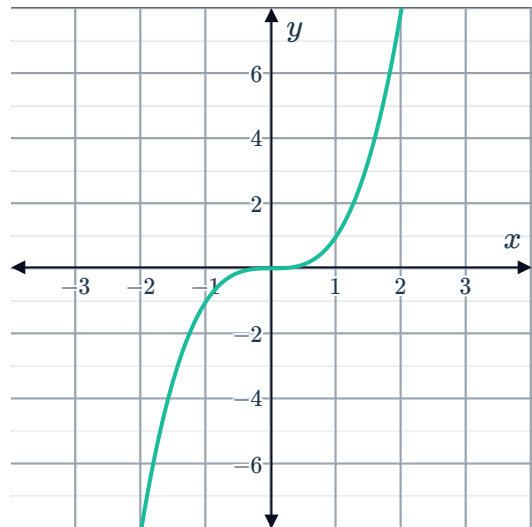
- a One of the points in the table is incorrect. Which point is it?
- b Bianca wants to find one other pair of values that satisfy $y = 20 - x^3$ before graphing the curve. Find the ordered pair when the x -coordinate is 6.
- c Plot the complete set of solutions for $y = 20 - x^3$, making sure that the curve goes through all points that satisfy it.

18 Consider the curve $y = -2x^3 + 16$.

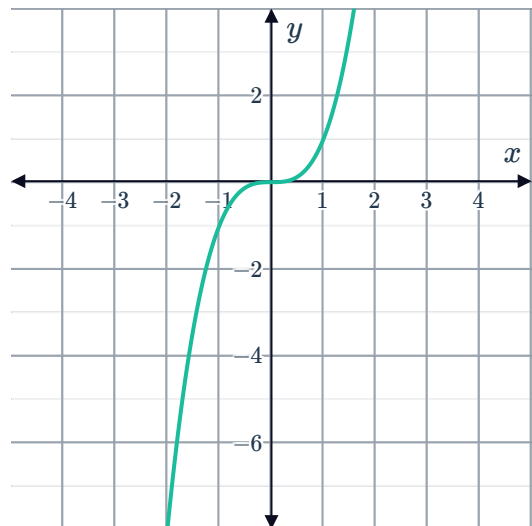


- a Find the x -intercept.
- b Find the y -intercept.
- c State the coordinates of the point of inflection.
- d Find the domain of the function in interval notation.
- e Find the range of the function in interval notation.
- f Sketch the graph of the curve.

19 A graph of $f(x) = x^3$ is shown. Sketch the curve after it has undergone transformations resulting in the function $g(x) = x^3 - 4$.



20 A graph of $y = x^3$ is shown. Sketch the curve after it has undergone transformations resulting in the function $y = 2(x - 2)^3 - 2$.



21 For the following cubic functions:



- i Determine whether the cubic is increasing or decreasing from left to right.
- ii Determine whether the cubic is more or less steep than the cubic $y = x^3$.
- iii Find the coordinates of the point of inflection of the cubic.
- iv Sketch the graph.

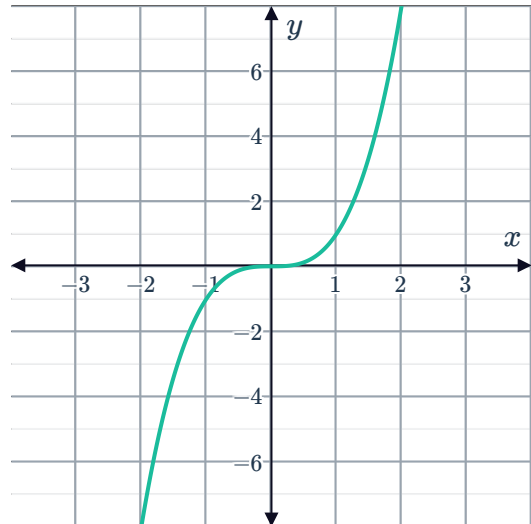
a $y = 2x^3 + 2$

b $y = 4x^3 - 3$

c $y = -\frac{x^3}{4} + 2$

22 Consider the graph of $y = x^3$ shown:

- a How do we shift the graph of $y = x^3$ to get the graph of $y = (x - 2)^3 - 3$?
- b Hence, sketch $y = (x - 2)^3 - 3$.



23 Consider the function $y = 2(x - 2)^3 - 2$

- a Is the cubic increasing or decreasing from left to right?
- b Is the function more or less steep than the function $y = x^3$?
- c What are the coordinates of the inflection point of the function?
- d Sketch the graph $y = 2(x - 2)^3 - 2$.

24 Consider the curve $y = 2(x + 1)^3 + 16$.

- a Find the x -intercept.
- b Find the y -intercept.
- c State the coordinates of the point of inflection.
- d Sketch the graph of the curve.

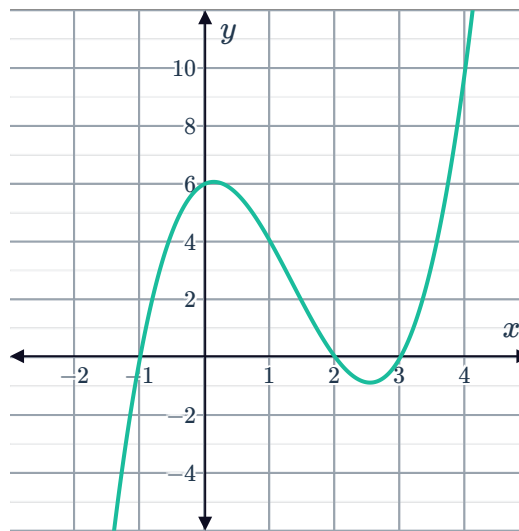
25 Consider the curve $y = -3(x - 1)^3 + 3$.



- a** Find the x -intercept.
- b** Solve for the y -intercept.
- c** State the coordinates of the point of inflection.
- d** Sketch the graph of the curve.

Factored form of a cubic function

26 The cubic function $y = (x + 1)(x - 2)(x - 3)$ has been graphed below. Determine the number of solutions to the equation $(x + 1)(x - 2)(x - 3) = 0$.



27 A cubic function has the equation $y = x(x - 4)(x - 3)$. How many x -intercepts will it have?

28 State whether the following functions pass through the origin:

- | | |
|--------------------------------------|--------------------------------|
| a $y = (x - 2)^2(x + 3)$ | b $y = (x + 1)^3$ |
| c $y = (x - 4)(x + 7)(x - 5)$ | d $y = x(x - 6)(x + 8)$ |

29 State whether each function has exactly two x -intercepts:

- | | |
|--------------------------------------|---------------------------------|
| a $y = (x + 3)^3$ | b $y = (x + 6)^2(x + 5)$ |
| c $y = (x + 7)(x - 1)(x - 4)$ | d $y = x(x - 2)(x - 8)$ |

30 Consider the cubic function $y = (x + 3)(x - 2)(x - 5)$.

- a** Determine the x -intercepts.
- b** A second cubic function has the same x -intercepts, but is a reflection of the above function about the y -axis. State the equation of the reflected function.



31 Sketch the function $y = (x - 2)(x + 1)(x + 4)$ showing the general shape of the curve and the x -intercepts.

32 Sketch the graph of $f(x) = (x - 2)^2(x - 4)$ and $g(x) = -(x - 2)^2(x - 4)$ on the same number plane.

33 Consider the function $y = (x + 3)^3$.

- a** Complete the following table of values: **b** Sketch the graph.

x	-5	-4	-3	-2	-1
y					

- c** State the domain. **d** State the range.

34 Consider the function $y = (x - 2)^3$.

- a** Complete the following table of values: **b** Sketch the graph.

x	0	1	2	3	4
y					

35 Consider the function $y = (x + 1)^3$.

- a** Complete the following table of values: **b** Sketch the graph.

x	-3	-2	-1	0	1
y					

36 Consider the equation $y = (x - 4)^3$.

- a** Complete the set of solutions for the above equation.
 $A(3, 1), B(2, 1), C(5, 1), D(4, 1), E(6, 1)$
- b** Plot the points on a coordinate axes.
- c** On the same axes, plot the curve that results from the entire set of solutions for the equation being graphed.
- d** Consider $x = 6.17$. According to the points on the graph, between which two integer values should the corresponding y -value lie?

37 For each of the functions below:



i Find the x -intercepts.

ii Find the y -intercept.

iii Sketch the graph.

a $y = (x + 3)(x + 2)(x - 2)$

b $y = -(x + 4)(x + 2)(x - 1)$

c $y = (x - 2)^2(x + 5)$

38 For each of the functions below:

i Express the equation in factorised form.

ii Find the y -intercept of the graph.

iii Find the x -intercepts of the graph.

iv Sketch the graph of the curve.

a $y = 3x + 2x^2 - x^3$

b $y = x^3 - 4x^2 - 7x + 10$

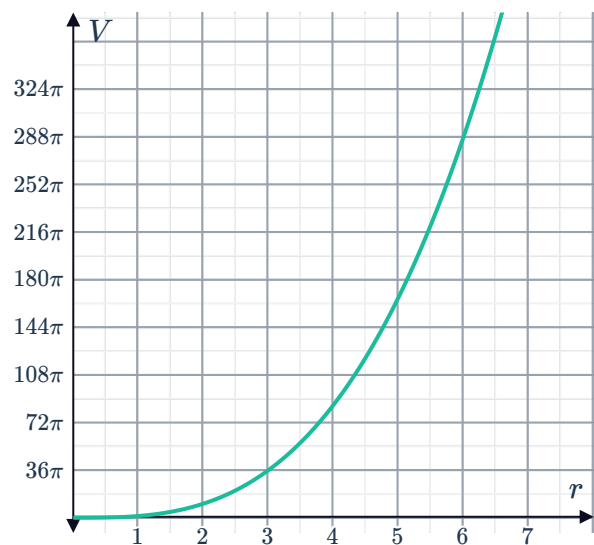
Applications

39 The volume of a sphere has the formula

$V = \frac{4}{3}\pi r^3$. The graph relating r and V is shown:

a Fill in the following table of values for the equation $V = \frac{4}{3}\pi r^3$, in terms of π :

r	1	2	4	5	7
V					



b Determine whether the following intervals show the volume, V of a sphere that has a radius measuring 4.5 m.

i $20\pi < V < 57\pi$

ii $85\pi < V < 166\pi$

iii $179\pi < V < 696\pi$

iv $221\pi < V < 366\pi$

c Using the graph, what is the radius of a sphere of volume $288\pi \text{ m}^3$?

40 A cube has side length 7 cm and a mass of  g. The mass of the cube is directly proportional to the cube of its side length.

- Let k be the constant of proportionality for the relationship between the side length x and the mass m of a cube. Find the value of k .
- Hence state the equation relating the mass (m) and side length (x) of a cube.
- Complete the table of values:

x	1	2	3	4
m				

- Sketch the graph of your equation.
- From the equation, find the mass of a cube with side 8.5 cm, to the nearest gram.
- A cube has a mass of 1920 g. From your graph, determine what whole number value its side length is closest to.

41 A box without a top cover is to be constructed from a rectangular cardboard, measuring 6 cm by 10 cm by cutting out four square corners of length x cm. Let V represent the volume of the box.

- Express the volume V of the box in terms of x , writing the equation in factorised form.
- For what range of values of x is the volume function defined?
- Sketch the graph of the volume function.
- Determine the volume of a box that has a height equivalent to the shorter dimension of the base.

42 A cylinder of radius x and height $2h$ is inscribed in a sphere of radius $\sqrt{15}$, centre at O as shown:

- Form an equation relating x and h .
- Form an expression for V , the volume of the cylinder, in terms of h .
- Determine the domain of h .
- The cylinder with the largest possible volume has a height of $2\sqrt{5}$, so $h = \sqrt{5}$. Determine the exact volume of this cylinder.

